Distributionally Robust LMI Control

Direct Data-Driven and DRC Synthesis

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Contents

1 Distributionally Robust LMI Synthesis for LTI Systems **2**
1.1 Problem Setting [1] 2
1.2 Paper-like Model Setting 3
1.3 Optimisation Problems [1] 4
1.4 Direct Data-Driven DRO–LMI via Young-Type Relaxation 5

A Utils **8**
A.1 Disturbance Evaluation 8
A.2 Medoid Selection 14

B Demonstrations **21**
B.1 Closed-loop lifting and decision variables 21
B.2 Post-optimization recovery of Δ_A and Δ_B 25

Bibliography **27**

Chapter 1

Distributionally Robust LMI Synthesis for LTI Systems

1.1 Problem Setting [1]

Output Feedback Control.

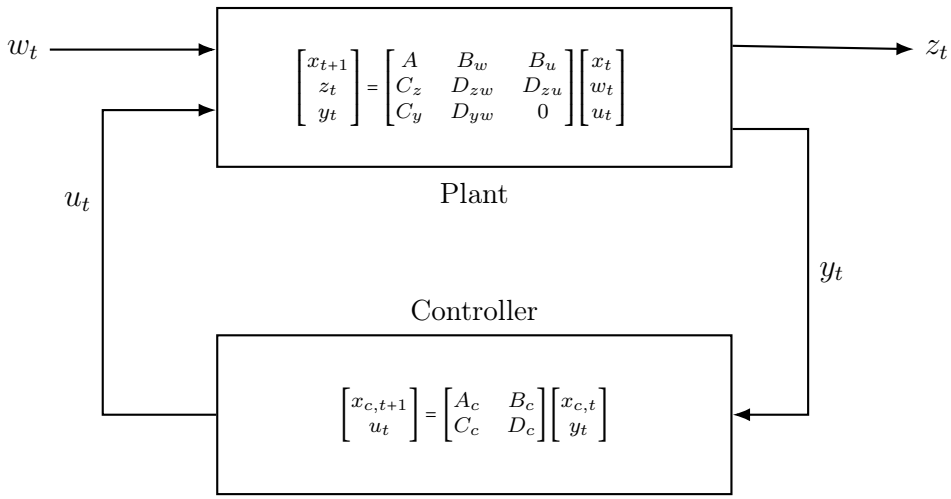


Figure 1.1: Closed-loop system with y feedback.

Define the combined state $\chi_t = [x_t^\top, x_{c,t}^\top]^\top$ and the closed-loop matrices

$$\mathcal{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix},$$
$$\mathcal{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad \mathcal{D} = D_{zw} + D_{zu} D_c D_{yw}.$$

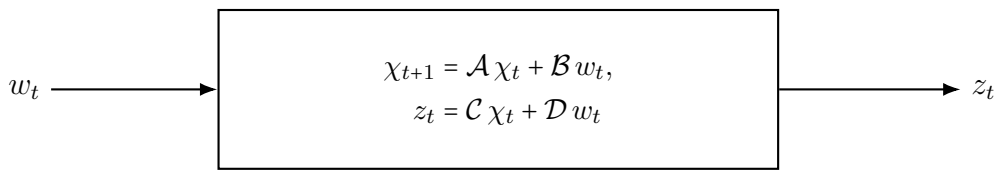


Figure 1.2: Lifted closed loop with state χ_t and matrices $(\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D})$.

The disturbance formulation is formulated in Appendix A.1.

1.2 Paper-like Model Setting

The *paper-like* configuration [2] refers to a linear time-invariant (LTI) system of the form:

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix},$$

where $x_t \in \mathbb{R}^{n_x}$ denotes the state, $u_t \in \mathbb{R}^{n_u}$ the control input, and $w_t \in \mathbb{R}^{n_w}$ an exogenous disturbance. The vectors $y_t \in \mathbb{R}^{n_y}$ and $z_t \in \mathbb{R}^{n_z}$ represent the measured and performance outputs, respectively.

The continuous-time version of this model describes a flexible turbine–shaft–pitch system:

$$\dot{x}(t) = A_{CT}x(t) + B_{CT}u(t) + E_{CT}w(t),$$

with

$$A_{CT} = \begin{bmatrix} \frac{k_\omega}{J} & \frac{k_h}{J} & 0 & \frac{k_\phi}{J} & 0 & \frac{k_\beta}{J} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{f_\omega}{m} & -\omega_f^2 & -2\zeta_f\omega_f & 0 & 0 & \frac{f_\beta}{m} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{m_\omega}{I_t} & 0 & 0 & -\omega_t^2 & -2\zeta_t\omega_t & \frac{m_\beta}{I_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega_p^2 & -2\zeta_p\omega_p \end{bmatrix},$$

$$B_{CT} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega_p^2 \end{bmatrix}, \quad E_{CT} = \begin{bmatrix} \frac{k_v}{J} & 0 \\ 0 & 0 \\ \frac{f_v}{m} & \frac{f_v}{m} \\ 0 & 0 \\ \frac{m_v}{I_t} & \frac{m_v}{I_t} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

which is discretized with sampling time T_s via

$$A = e^{A_{CT}T_s}, \qquad B_u = \int_0^{T_s} e^{A_{CT}\tau} B_{CT}\, d\tau, \qquad B_w = \int_0^{T_s} e^{A_{CT}\tau} E_{CT}\, d\tau.$$

The output matrices $(C_z, D_{zw}, D_{zu}, C_y, D_{yw})$ are selected to reflect a standard $\mathcal{H}_2$-type performance objective:

$$z_t = \begin{bmatrix} Q_x^{1/2} & 0 \\ 0 & R_u^{1/2} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \qquad Q_x \succeq 0, R_u \succ 0 \qquad y_t = C_y x_t + D_{yw} w_t,$$

with $Q_x \succeq 0, R_u \succ 0$ and $D_{zw} = 0$.

1.3 Optimisation Problems [1]

Baseline

$$\min_{\theta \leftrightarrow (A_c, B_c, C_c, D_c)} J(\theta), \qquad \text{s.t. } \mathcal{A} \quad \text{isstable}$$

$$J(\theta) = \sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \, \|z_t\|^2 \tag{P1}$$

$$\text{s.t.} \quad \begin{aligned} \chi_{t+1} &= \mathcal{A} \chi_t + \mathcal{B} w_t, \\ z_t &= \mathcal{C} \chi_t + \mathcal{D} w_t \end{aligned}, \quad \chi_0 = 0$$

LMI

Introducing decision variables

$$\lambda \geq 0, \quad Q, \quad X = X^\top, \; Y = Y^\top, \; K, L, M, N,$$

and defining

$$\mathbb{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \quad \mathbf{D} = D_{zw} + D_{zu} N D_{yw}.$$

The objective bacame:

$$\min_{\lambda \geq 0, \; Q, \; X = X^\top, \; Y = Y^\top, \; K, L, M, N} \quad \operatorname{trace}(Q \Sigma_{\text{nom}}) + \lambda \gamma^2 \tag{P2}$$

subject to a block LMI that depends on the ambiguity model.

Correlated model ($\mathcal{W}_{\text{cor}}$). With identity blocks I_w, I_z of matching sizes, enforce

$$\begin{bmatrix} -\mathbb{P} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

Independent model ($\mathcal{W}_{\text{ind}}$). Impose the pair of LMIs

$$\begin{bmatrix} -\mathbb{P} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -\mathbb{P} & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \qquad \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

1.4 Direct Data-Driven DRO–LMI via Young-Type Relaxation

Idea. We first estimate the nominal plant maps from one persistently exciting trajectory $\{x_k, u_k, y_k, z_k\}_{k=0}^{T}$ by one–step regression. The remaining data mismatch is treated as an *uncertainty budget* and injected into the robust LMI through a Young-inequality relaxation. This keeps the synthesis convex while avoiding an explicit probabilistic model of the identification errors.

One–step estimates and residual decomposition

Form the stacked data matrices, collected as shown in Appendix A.2

$$X_- = \begin{bmatrix} x_0 & \cdots & x_{T-1} \end{bmatrix}, \qquad U_- = \begin{bmatrix} u_0 & \cdots & u_{T-1} \end{bmatrix}, \qquad X_+ = \begin{bmatrix} x_1 & \cdots & x_T \end{bmatrix}.$$

$$Y_- = \begin{bmatrix} y_0 & \cdots & y_{T-1} \end{bmatrix}, \qquad Z_- = \begin{bmatrix} z_0 & \cdots & z_{T-1} \end{bmatrix},$$

SysID Define $D_x = \begin{bmatrix} X_- \\ U_- \end{bmatrix}$ and use the (regularized) right inverse $D_x^\dagger := D_x^\top (D_x D_x^\top + \rho I)^{-1}$ (with $\rho \geq 0$, $\rho = 10^{-6}$ in our implementation). The nominal state update is

$$\underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} = X_+ D_x^\dagger, \qquad \hat{A} \in \mathbb{R}^{n\times n},\ \hat{B}_u \in \mathbb{R}^{n\times m}.$$

Let the *state residuals* be

$$R := X_+ - \hat{A} X_- - \hat{B}_u U_- \in \mathbb{R}^{n\times T}.$$

We extract a low–rank disturbance channel from the principal directions of the residual covariance

$$S_R := \frac{1}{T} R R^\top, \qquad \tilde{S}_R := \tfrac{1}{2}\big(S_R + S_R^\top\big) + \varepsilon I_n = U_s \operatorname{diag}(s_1, \ldots, s_n)\, U_s^\top,$$

with eigenvalues ordered as $s_1 \geq \cdots \geq s_n \geq 0$.

Data–driven covariance modes. When the disturbance covariance is not known a priori, we choose the minimal rank n_w such that

$$n_w := \min\Big\{k : \sum_{i=1}^{k} s_i \ \geq\ \eta \sum_{i=1}^{n} s_i\Big\},$$

and define

$$\hat{B}_w \widehat{\Sigma}_w^{1/2} := U_s(:, 1{:}n_w) \operatorname{diag}\big(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}\big), \qquad \hat{W} := \hat{B}_w^\dagger R,$$

with $\widehat{\Sigma}_w := \operatorname{diag}(s_1, \ldots, s_{n_w})$ and nominal disturbance covariance

$$\Sigma_{\text{nom}} := \tfrac{1}{2}\Big(\tfrac{1}{T}\hat{W}\hat{W}^\top + \big(\tfrac{1}{T}\hat{W}\hat{W}^\top\big)^\top\Big) + \varepsilon I_{n_w}.$$

Known covariance mode. When a nominal disturbance covariance $\Sigma_{\text{nom}} \succ 0$ is known a priori, with dimension $n_w \times n_w$, we fix

$$n_w := \dim(\Sigma_{\text{nom}}).$$

Let

$$\Sigma_{\text{nom}} = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_{n_w})\,Q^\top, \qquad \Sigma_{\text{nom}}^{-1/2} := Q\,\mathrm{diag}(\lambda_1^{-1/2}, \ldots, \lambda_{n_w}^{-1/2})\,Q^\top,$$

and denote by $U_p := U_s(:, 1:n_w)$ the eigenvectors of $\tilde{S}_R$ associated with the largest n_w eigenvalues $(s_1, \ldots, s_{n_w})$. We then set

$$\hat{B}_w := U_p\,\mathrm{diag}\big(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}\big)\,\Sigma_{\text{nom}}^{-1/2}.$$

By construction,

$$\hat{B}_w\,\Sigma_{\text{nom}}\,\hat{B}_w^\top = U_p\,\mathrm{diag}(s_1, \ldots, s_{n_w})\,U_p^\top,$$

so the dominant part of the residual covariance is matched using the prescribed disturbance covariance Σ_{nom}, and no additional re-estimation of Σ_{nom} from R is performed in this mode.

For the performance outputs we fix the weighting matrices as in the reference setup, defining $z = [\,Q_x^{1/2}x;\ R_u^{1/2}u\,]$ with

$$C_z = \begin{bmatrix} \mathrm{diag}(\sqrt{Q_x}) \\ 0_{n_u \times n_x} \end{bmatrix}, \quad D_{zu} = \begin{bmatrix} 0_{n_x \times n_u} \\ \mathrm{diag}(\sqrt{R_u}) \end{bmatrix}, \quad D_{zw} = 0, \quad Q_x = I_{n_x},\ R_u = I_{n_u}$$

so that the cost $\|z\|_2^2$ is consistent across the BL and DD formulations—this is a deliberate designer choice for fair comparison.

A data–driven radius for model error

We bound the *identification error* on the state map with $\Delta_x \le RD^\dagger = \beta$ from $\hat{O}_x = X_+D^\dagger := (O_xD)D^\dagger + RD^\dagger = O_x + \Delta_x$.

Here Δ_x collects the stacked deviations in $[A\ B_u]$ relative to $\hat{O}_x$. This bound is conservative but computable directly from data.

$$\begin{bmatrix} A & B_u \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} + \Delta_x, \quad \Delta_x = \begin{bmatrix} \Delta_A & \Delta_B \end{bmatrix}, \quad \|\Delta_i\|_F \le \beta_i$$

with

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x + n_u}}, \quad \beta_B = \beta\sqrt{\frac{n_u}{n_x + n_u}}$$

Therefore, writing the perturbed lifted block as $\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E$, with $E \prec \beta_E$. Formal derivations and the precise kernel substitutions are given in Appendix B.1; the post–optimization reconstruction of the nominal maps from the recovered errors, $A \leftarrow \hat{A} + \Delta_A$, $B_u \leftarrow \hat{B}_u + \Delta_B$, is detailed in Appendix B.2.

Final program. The Young-relaxed data-driven synthesis (correlated case) is

$$\min_{\substack{\lambda\geq 0,\ Q\ \geq\ 0,\\ S\ \geq\ 0\ \ P_s\ \geq\ 0\, H\ \geq\ 0\ \ G\ =\ \sigma_p I\ =\ G^T\ \geq\ 0\\ \sigma_s\ \geq\ 0,\ \sigma_p\ \geq\ 0,\ \{t\}_{i\in(K,L,M,N,X,Y)}\\ X{=}X^T,\ Y{=}Y^T,\ K,\ L,\ M,\ N}} \quad \mathrm{tr}(Q\,\Sigma_{\mathrm{nom}}) + \lambda\gamma^2 \ + \sum_{i\in(K,L,M,N,X,Y)} \rho\,(t_i + \varepsilon)$$

$$\begin{aligned}
\text{s.t.}\ \ & P = P^T = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \quad \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) \ \prec\ 0 \\
& \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{1,1} \leftarrow -P \ +\ (\sigma_s + \sigma_p)\ \beta_E \\
& \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{n,n} \leftarrow -P_s \\
& \begin{bmatrix} G & I \\ I & P \end{bmatrix} \succ 0, \qquad \begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0, \qquad \begin{bmatrix} G+H & I \\ I & P_s \end{bmatrix} \succ 0 \\
& \sigma_s I - S \succ 0\ , \quad \begin{bmatrix} t_j I_{n^j_{row}} & j \\ j^\top & t_j I_{n^j_{col}} \end{bmatrix}\Bigg|_{j\in(K,L,M,N,X,Y)} \geq 0.
\end{aligned}$$

where $\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}$ are built from the nominal data-based maps $(\hat{A},\hat{B},\hat{C}_y,\hat{D}_{yw},\hat{C}_z,\hat{D}_{zu},\hat{D}_{zw})$. For the correlated case, we have that $n = 4$, whereas in the independent one $n = 2$.

Appendix A

Utils

A.1 Disturbance Evaluation

The disturbance sequences w_t used in both the baseline and distributionally robust simulations are drawn from a *Wasserstein ambiguity set* [3, 4] centered at a nominal Gaussian model. The ambiguity set captures distributional uncertainty in the second-order statistics of the process noise, rather than assuming a fixed covariance. We fix a nominal zero-mean Gaussian distribution

$$P_{\text{nom}} = \mathcal{N}(0, \Sigma_{\text{nom}}), \qquad \Sigma_{\text{nom}} \in \mathbb{S}^{n_w}_{++},$$

and consider all zero-mean Gaussians whose covariance remains within a prescribed distance of Σ_{nom} in the 2-Wasserstein (Gelbrich) metric.

For any zero-mean Gaussians $P_1 = \mathcal{N}(0, \Sigma_1),\ P_2 = \mathcal{N}(0, \Sigma_2)$, the squared 2-Wasserstein distance is

$$W_2^2(P_1, P_2) = \operatorname{tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2})^{1/2}),$$

or, equivalently,

$$W_2^2(\mathcal{N}(0, S_1), \mathcal{N}(0, S_2)) = \operatorname{tr}(S_1) + \operatorname{tr}(S_2) - 2\operatorname{tr}\Big[\big(S_2^{1/2}S_1S_2^{1/2}\big)^{1/2}\Big].$$

The corresponding closed Wasserstein ball of radius $\gamma > 0$ around P_{nom} is

$$\mathcal{B}_\gamma = \Big\{\mathcal{N}(0, \Sigma) : \Sigma \in \mathbb{S}^{n_w}_{++},\ W_2\big(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})\big) \le \gamma\Big\}.$$

Equivalently, in covariance form, we define the 2-Wasserstein ambiguity set as

$$\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) = \Big\{\mathcal{N}(0, \Sigma) \;\Big|\; W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \le \gamma\Big\},$$

so that the set represents all Gaussian distributions whose covariance lies within distance γ (in the Bures metric) from the nominal covariance Σ_{nom}. All sampled disturbances are zero-mean Gaussians with covariance $\Sigma_{\text{use}} \in \mathcal{B}_\gamma$, and hence lie in this ambiguity set.

Given a candidate covariance $\Sigma_{\text{raw}} \in \mathbb{S}^{n_w}_{++}$, we first check its distance to the nominal model:

$$d_{\text{raw}} = W_2(\mathcal{N}(0, \Sigma_{\text{raw}}), \mathcal{N}(0, \Sigma_{\text{nom}}))\,.$$

If $d_{\text{raw}} \le \gamma$, we simply keep $\Sigma_{\text{use}} = \Sigma_{\text{raw}}$. Otherwise, we move from Σ_{raw} towards Σ_{nom} along the Bures–Wasserstein geodesic connecting them. Let

$$S = \Sigma_{\text{raw}}^{1/2}, \qquad A = \big(S\Sigma_{\text{nom}}S\big)^{1/2},$$

and define the geodesic

$$\Sigma(t) = \big((1-t)S + tA\big)^2, \qquad t \in [0,1].$$

We then choose $t^\star \in (0,1]$ such that

$$W_2(\mathcal{N}(0,\Sigma(t^\star)), \mathcal{N}(0,\Sigma_{\text{nom}})) = \gamma,$$

and set

$$\Sigma_{\text{use}} := \Sigma(t^\star).$$

In the actual implementation, $t^\star$ is found by a scalar bisection search on t. Thus, every covariance used for sampling either lies strictly inside the ball $d_{\text{raw}} \le \gamma$ or is obtained by projecting a raw candidate onto the boundary of $\mathcal{B}_\gamma$ along the Bures geodesic.

Two simulation modes are available, differing only in their temporal structure while remaining consistent with the same marginal ambiguity model.

In the *independent (iid)* case, given $\Sigma_{\text{use}} \in \mathcal{B}_\gamma$ and time horizon T, we generate an i.i.d. disturbance sequence

$$w_t \sim \mathcal{N}(0,\Sigma_{\text{use}}), \qquad t = 0,\dots,T-1,$$

independent across t. Equivalently, if L is a Cholesky factor $\Sigma_{\text{use}} = LL^\top$ and $z_t \sim \mathcal{N}(0, I_{n_w})$, then

$$w_t = Lz_t, \qquad t = 0,\dots,T-1.$$

This sequence belongs to the i.i.d. ambiguity set $\mathcal{W}_{\text{ind}}$ and corresponds to temporally uncorrelated samples inside the Wasserstein ball. When we specialize to the nominal covariance, this reduces to

$$w_t \sim \mathcal{N}(0,\Sigma_{\text{nom}}), \quad t = 0,\dots,T-1,$$

with implementation via a Cholesky factorization $L = \text{chol}(\Sigma_{\text{nom}})$ and $w_t = L\,z_t,\ z_t \sim \mathcal{N}(0,I)$.

In the *correlated* case, $\mathcal{W}_{\text{cor}}$, we obtain temporally correlated disturbances with stationary marginal $\mathcal{N}(0,\Sigma_{\text{use}})$ via an AR(1) model

$$w_{t+1} = Fw_t + \varepsilon_t, \qquad t = 0,\dots,T-2,$$

with

$$F = \rho I_{n_w}, \qquad \varepsilon_t \sim \mathcal{N}(0,Q),$$

where $|\rho| < 1$ ensures stability and Q is chosen so that the stationary covariance equals Σ_{use}. This is enforced by the discrete-time Lyapunov equation

$$\Sigma_{\text{use}} = F\Sigma_{\text{use}}F^\top + Q,$$

i.e.

$$Q = \Sigma_{\text{use}} - F\Sigma_{\text{use}}F^\top.$$

We initialize the process in stationarity,

$$w_0 \sim \mathcal{N}(0,\Sigma_{\text{use}}),$$

and then iterate the AR(1) recursion. The resulting trajectory $\{w_t\}_{t=0}^{T-1}$ has marginal distribution $\mathcal{N}(0, \Sigma_{\text{use}})$ at every t. A specialized instance preserving a nominal covariance Σ_{nom} uses

$$w_{t+1} = \rho\, w_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$

with $0 < \rho < 1$ and

$$\Sigma_{\text{nom}} = \rho^2 \Sigma_{\text{nom}} + Q \quad \Rightarrow \quad Q = (1 - \rho^2)\Sigma_{\text{nom}}.$$

This yields stationary correlated samples having marginal covariance Σ_{nom}, ensuring consistency with the ambiguity model.

Operationally, the sampling procedure can be described as follows. We require a nominal covariance $\Sigma_{\text{nom}} \in \mathbb{S}_{++}^{n_w}$, a radius $\gamma > 0$, a time horizon T, a mode $\texttt{mode} \in \{\text{independent}, \text{correlated}\}$, and a correlation coefficient $\rho \in (0, 1)$. We choose a candidate covariance $\Sigma_{\text{raw}} \in \mathbb{S}_{++}^{n_w}$, for instance

$$\Sigma_{\text{raw}} = \begin{bmatrix} 0.69 & 0.53 \\ 0.53 & 1.61 \end{bmatrix},$$

and compute

$$d_{\text{raw}} = W_2(\mathcal{N}(0, \Sigma_{\text{raw}}), \mathcal{N}(0, \Sigma_{\text{nom}}))\,.$$

If $d_{\text{raw}} \le \gamma$ we set $\Sigma_{\text{use}} = \Sigma_{\text{raw}}$. Otherwise we compute $S = \Sigma_{\text{raw}}^{1/2}$ and $A = (S\Sigma_{\text{nom}}S)^{1/2}$, define the geodesic $\Sigma(t) = ((1-t)S + tA)^2$ for $t \in [0, 1]$, and find $t^\star \in (0, 1]$ by bisection such that

$$W_2(\mathcal{N}(0, \Sigma(t^\star)), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma,$$

then set $\Sigma_{\text{use}} = \Sigma(t^\star)$.

If the mode is independent, we compute a Cholesky factor $\Sigma_{\text{use}} = LL^\top$ and, for $t = 0, \ldots, T-1$, draw $z_t \sim \mathcal{N}(0, I_{n_w})$ and set

$$w_t = Lz_t.$$

If the mode is correlated, we set $F = \rho I_{n_w}$ and

$$Q = \Sigma_{\text{use}} - F\Sigma_{\text{use}}F^\top,$$

compute Cholesky factors $\Sigma_{\text{use}} = L_\Sigma L_\Sigma^\top$ and $Q = L_Q L_Q^\top$, draw $z_0 \sim \mathcal{N}(0, I_{n_w})$ and set

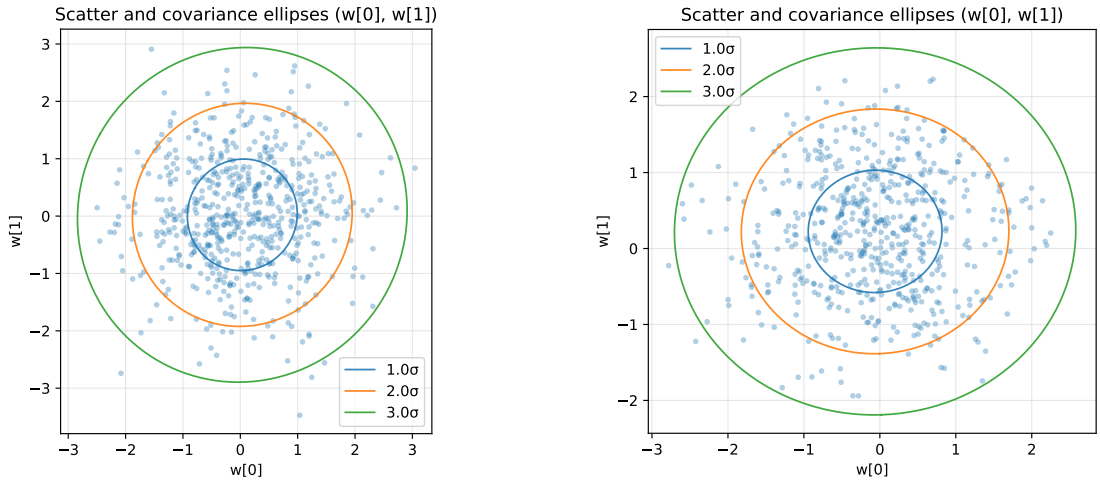
$$w_0 = L_\Sigma z_0,$$

and for each $t = 0, \ldots, T-2$, draw $\eta_t \sim \mathcal{N}(0, I_{n_w})$, set $\varepsilon_t = L_Q \eta_t$ and update

$$w_{t+1} = Fw_t + \varepsilon_t.$$

The resulting sequence $\{w_t\}_{t=0}^{T-1}$ is then used as the disturbance realization in simulation.

Each figure below compares the *independent (iid)* and the *correlated (AR(1))* disturbance samplers that both lie in a 2-Wasserstein ball of radius γ around $\mathcal{N}(0, \Sigma_{\text{nom}})$. The iid and AR(1) samplers share the same stationary marginal $\mathcal{N}(0, \Sigma_{\text{nom}})$; they differ only in temporal structure (autocorrelation).



(a) Independent (iid): two-dimensional marginal samples (w_i, w_j) with nominal and empirical covariance ellipses. The empirical ellipse matches the nominal, confirming $\mathrm{Cov}(w_t) = \Sigma_{\mathrm{nom}}$ under iid sampling.

(b) Correlated (AR(1)): same marginal visualization. Marginals agree with Σ_{nom}; differences from iid are temporal, not marginal.

Figure A.1: Scatter and covariance ellipses. The panels visualize *marginal* covariance in the plane (w_i, w_j).

Both modes satisfy $\mathbb{E}[w_t] = 0$ and $\mathrm{Cov}(w_t) = \Sigma_{\mathrm{nom}}$, hence the nearly identical scatter/ellipse and PDF panels. The difference is temporal:

$$\text{iid:} \quad \mathrm{Cov}(w_{t+k}, w_t) = 0 \ (k \neq 0), \qquad \text{AR(1):} \quad \mathrm{Cov}(w_{t+k}, w_t) = \rho^{|k|}\Sigma_{\mathrm{nom}}.$$

This is invisible in time-agnostic marginals. To expose it, one can add time-series diagnostics such as a trace of $w_i(t)$, an ACF plot (iid: spike at lag 0; AR(1): geometric decay ρ^k), or a lag-1 scatter (w_t, w_{t+1}) that aligns along the 45° line when correlated.

A technical caveat concerns the construction of boundary ellipses. The boundary covariances are constructed by eigenvalue shifts along principal axes of Σ_{nom}. This is exact in the commuting case (shared eigenvectors) and remains a useful visual surrogate otherwise; it illustrates the scale of admissible covariance perturbations at radius γ in the Bures/W_2 metric.

Elliptical covariance visualization

For any zero-mean Gaussian distribution $\mathcal{N}(0, \Sigma) \subset \mathbb{R}^{n_w}$, its 2-dimensional marginal over coordinates (i, j) has covariance

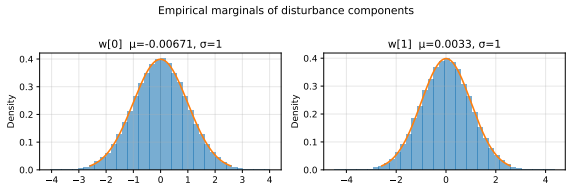
$$\Sigma_{ij} := \begin{bmatrix} \Sigma_{ii} & \Sigma_{ij} \\ \Sigma_{ji} & \Sigma_{jj} \end{bmatrix} \in \mathbb{S}^2_{++}.$$

The associated *k-sigma covariance ellipse* is the set

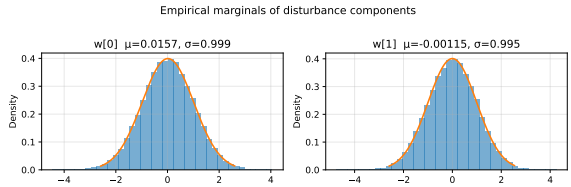
$$\mathcal{E}_k(\Sigma_{ij}) = \left\{ \mu_{ij} + V\Lambda^{1/2}Q : Q \in \mathbb{R}^2, \ \|Q\| \leq k \right\},$$

where

$$\Sigma_{ij} = V\Lambda V^{\top},$$



(a) Independent (iid): empirical marginal densities of disturbance components. Means are near zero and variances match the diagonal of Σ_{nom}.



(b) Correlated (AR(1)): empirical marginal densities. As expected, they match the iid case; correlation does not alter stationary marginals.

Figure A.2: One-step marginal PDFs. Time order is ignored, so iid and AR(1) look nearly identical at the marginal level, as $T \to \inf$.

$\Lambda = \operatorname{diag}(\lambda_1, \lambda_2)$, $V = [v_1\ v_2]$ contains orthonormal eigenvectors, and k corresponds to the chosen confidence radius (for instance $k = 2$ as a common visualization level). Geometrically, v_1 and v_2 give the orientation of the ellipse, $\sqrt{\lambda_1}$ and $\sqrt{\lambda_2}$ give the axis lengths, and μ_{ij} is the center (empirical sample mean).

Given sampled disturbances $w(t) \in \mathbb{R}^{n_w}$, the ellipse plots display the sample scatter $(w_i(t), w_j(t))$ for all t to visualize the empirical marginal distribution, the *nominal ellipse* obtained by projecting the nominal covariance Σ_{nom} to coordinates (i, j), the *empirical ellipse* corresponding to the sample covariance of the points $(w_i(t), w_j(t))$, and a family of *Wasserstein-boundary ellipses* obtained from synthetic covariances Σ_b that satisfy

$$W_2\big(\mathcal{N}(0, \Sigma_b), \mathcal{N}(0, \Sigma_{\text{nom}})\big) = \gamma.$$

These boundary covariances are created by modifying eigenvalues of Σ_{nom} by $\pm\gamma$ along each principal axis, and they visualize directions of maximal allowed covariance perturbation.

To generate these axis-aligned boundary covariances, we write

$$\Sigma_{\text{nom}} = U \Lambda U^\top \quad \text{with} \quad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_{n_w}),\ \lambda_k > 0,$$

and define the nominal square roots

$$s_k = \sqrt{\lambda_k}.$$

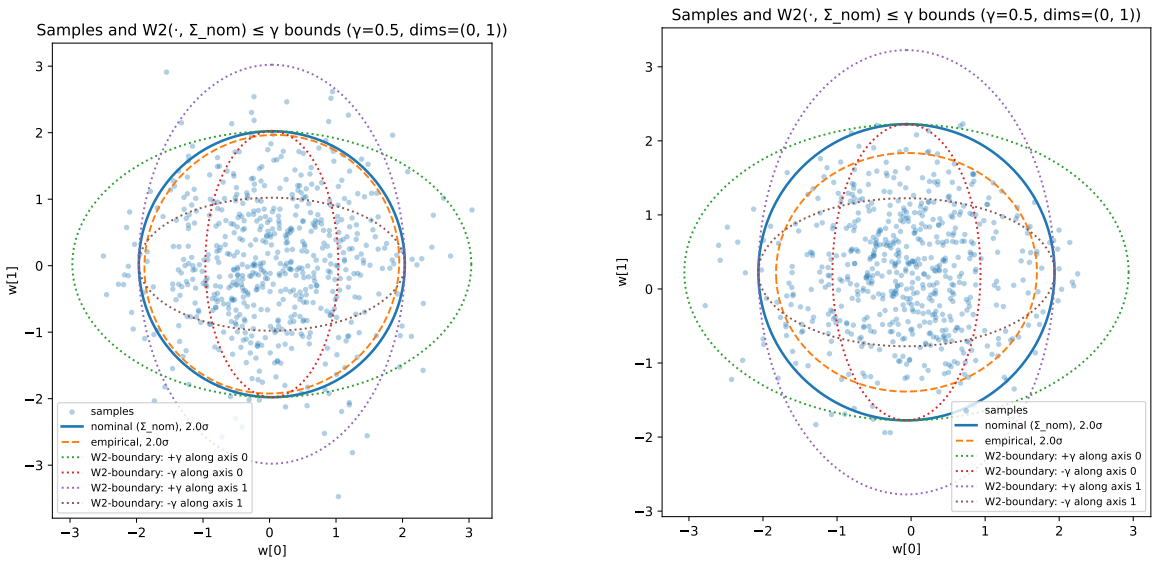
To construct illustrative points on the Wasserstein boundary, we modify a single axis at a time via

$$s'_k = \max(0,\ s_k \pm \gamma), \qquad \lambda'_k = (s'_k)^2,$$

and keep all other eigenvalues unchanged. Each boundary covariance is then

$$\Sigma_b = U \begin{bmatrix} \lambda'_1 & & \\ & \ddots & \\ & & \lambda'_{n_w} \end{bmatrix} U^\top,$$

and, in the commutative-eigenvector case, satisfies $W_2(\mathcal{N}(0, \Sigma_b), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma$. These matrices, when projected to the (i, j)-coordinates, generate the dotted ellipses plotted together with the nominal and empirical ones.



(a) Independent (iid): samples with nominal, empirical, and representative W_2-boundary ellipses. The nominal and empirical ellipses lie within the ball $W_2(\cdot, \Sigma_{\text{nom}}) \leq \gamma$.

(b) Correlated (AR(1)): correlated samples with the same W_2-boundary overlays. Correlation affects lag structure, not the W_2-marginal bounds.

Figure A.3: Wasserstein-ball overlays. Dotted ellipses illustrate covariances on the W_2 boundary around Σ_{nom}. They bound admissible *marginal* covariances, not individual points.

Finally, the plotting routine for marginal ellipses and Wasserstein bounds can be described explicitly in terms of the same objects. Given samples $w \in \mathbb{R}^{T \times n_w}$, a nominal covariance Σ_{nom}, a Wasserstein radius γ, a coordinate pair (i, j) and a confidence level k, we extract the 2D samples

$$W = w[:, (i, j)],$$

compute the empirical mean μ and sample covariance Σ_{emp}, and form the 2D nominal covariance

$$\Sigma_{\text{nom},2d} = \Sigma_{\text{nom}}[(i, j), (i, j)].$$

We then generate a small set of boundary covariances $\{\Sigma_b\}$ by axis-wise eigenvalue shifts of Σ_{nom} by $\pm\gamma$ as described above, and project each Σ_b to its (i, j) marginal. For each covariance

$$\Sigma \in \left\{\Sigma_{\text{nom},2d}, \Sigma_{\text{emp},2d}, \Sigma_b^{(i,j)}\right\},$$

we compute the eigen-decomposition

$$\Sigma = V \Lambda V^{\top},$$

and generate the ellipse points as

$$\mathcal{E}_k = \left\{\mu + V \Lambda^{1/2} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} k : \theta \in [0, 2\pi]\right\}.$$

The nominal ellipse is plotted as a solid line, the empirical ellipse as a dashed line and the boundary ellipses as dotted lines, while the sample points W are shown as a scatter

cloud. In this way, the plots jointly show the empirical marginal distribution, the nominal covariance, and synthetic covariances lying on the Wasserstein boundary.

In summary, the disturbances in simulation are not arbitrary noise sequences but samples from a *Wasserstein-consistent* uncertainty model. The constructions above ensure that both iid and correlated disturbance sequences have zero mean and covariance matrices belonging to a Wasserstein ball of radius γ around Σ_{nom}. This maintains coherence between the data-driven identification, the LMI synthesis and the subsequent Monte Carlo evaluation of closed-loop robustness.

A.2 Medoid Selection

Dataset preprocessing. Rather than relying on a single open-loop experiment, which may lead to different simulation results, multiple trajectories $\{\mathcal{T}_i\}_{i=1}^{N=50}$ are collected, each containing matrices

$$\mathcal{T}_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\}, \qquad X_i \in \mathbb{R}^{n_x \times T_i},\ U_i \in \mathbb{R}^{n_u \times T_i},\ Y_i \in \mathbb{R}^{n_y \times T_i},\ Z_i \in \mathbb{R}^{n_z \times T_i},$$

all truncated to a common horizon $T = \min_i T_i$. Each dataset is initialized with random states (i.e. $x_0 \sim \mathcal{N}(0, I)$) to ensure excitation diversity, and only trajectories satisfying the persistent–excitation condition $\sigma_{\min}\big(\frac{1}{T}\Phi\Phi^\top\big) > \varepsilon_{\text{PE}}$ are kept before medoid selection.

Before selecting a nominal dataset, the trajectories are *normalized* to prevent one signal from dominating the others. For each variable $k \in \{X, U, Y, Z, X_+\}$, a scale factor s_k is computed as the mean Frobenius norm across seeds,

$$s_k = \frac{1}{N} \sum_{i=1}^{N} \|\mathcal{T}_i[k]\|_F + \varepsilon,$$

and all data are rescaled as $\tilde{\mathcal{T}}_i[k] = \mathcal{T}_i[k]/s_k$. This ensures that signals with different physical units contribute comparably to the distance metric.

Among the N realizations, the most representative one (the *medoid*) is selected as

$$i^\star = \arg\min_i \sum_k w_k \sum_{j=1}^{N} \left\|\tilde{\mathcal{T}}_j[k] - \tilde{\mathcal{T}}_i[k]\right\|_F^2,$$

where w_k are optional weighting coefficients. The final matrices used for identification are then taken from this single dataset,

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+}).$$

Motivation. A naive arithmetic average of the trajectories, e.g. $\bar{X} = \frac{1}{N}\sum_i X_i$, is undesirable because opposite sign components can cancel, artificially shrinking the state amplitudes and destroying the dynamical consistency between X_- and X_+. The medoid procedure avoids this by selecting one real, physically consistent trajectory that is closest (in the Frobenius sense) to all others, effectively reducing the influence of measurement noise and random variations without corrupting the system structure.

Algorithm 1 Dataset Preprocessing and Medoid Selection

Input: N trajectories $\mathcal{T}_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\}$, time lengths T_i
Parameters: PE threshold $\varepsilon_{\mathrm{PE}}$, scaling offset ε, weights w_k
(A) Truncate and PE filtering
$T \leftarrow \min_i T_i$
for $i = 1$ to N **do**
    Truncate each $\mathcal{T}_i$ to horizon T
    Sample $x_0^{(i)} \sim \mathcal{N}(0, I)$
    Form $\Phi_i = [X_i; U_i]$
    if $\sigma_{\min}(\frac{1}{T}\Phi_i\Phi_i^\top) \le \varepsilon_{\mathrm{PE}}$ **then**
        discard $\mathcal{T}_i$
(B) Normalization across datasets
for each block $k \in \{X, U, Y, Z, X_+\}$ **do**
    Compute scale $s_k = \frac{1}{N}\sum_{i=1}^{N} \|\mathcal{T}_i[k]\|_F + \varepsilon$
    for $i = 1$ to N **do**
        $\tilde{\mathcal{T}}_i[k] \leftarrow \mathcal{T}_i[k]/s_k$
(C) Medoid selection
for $i = 1$ to N **do**
    Compute total normalized Frobenius distance:

$$d_i = \sum_k w_k \sum_{j=1}^{N} \|\tilde{\mathcal{T}}_j[k] - \tilde{\mathcal{T}}_i[k]\|_F^2$$

$i^\star = \arg\min_i d_i$
(D) Output
Representative dataset:

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+})$$

$\mathbf{N_{sims}}$ sweep and closed-loop metrics. The function `NsimSweep_FROM_DATA` systematically studies how the number of datasets N_{sims} used in the medoid selection affects the closed-loop performance of the distributionally robust controller. For each N_{sims} in a prescribed grid, the routine repeatedly calls the main synthesis-and-evaluation pipeline with `FROM_DATA=True`, fixed Wasserstein radius γ and fixed disturbance model, and collects three scalar quantities:

- the closed-loop performance cost J,
- the spectral radius $\rho(A_{\mathrm{cl}})$ of the closed-loop matrix,
- the optimization objective `obj` returned by the LMI solver.

For each N_{sims}, the function performs `runs_per_N` Monte Carlo repetitions and stores the empirical mean and standard deviation of $(J, \rho, \texttt{obj})$. After the sweep, it selects "best" values of N_{sims} for each metric via a scalar ranking: small mean and small standard deviation are preferred for J and `obj`, whereas for the spectral radius only stable designs with $\rho(A_{\mathrm{cl}}) < 1$ are considered and the smallest mean ρ (again with small variance) is selected.

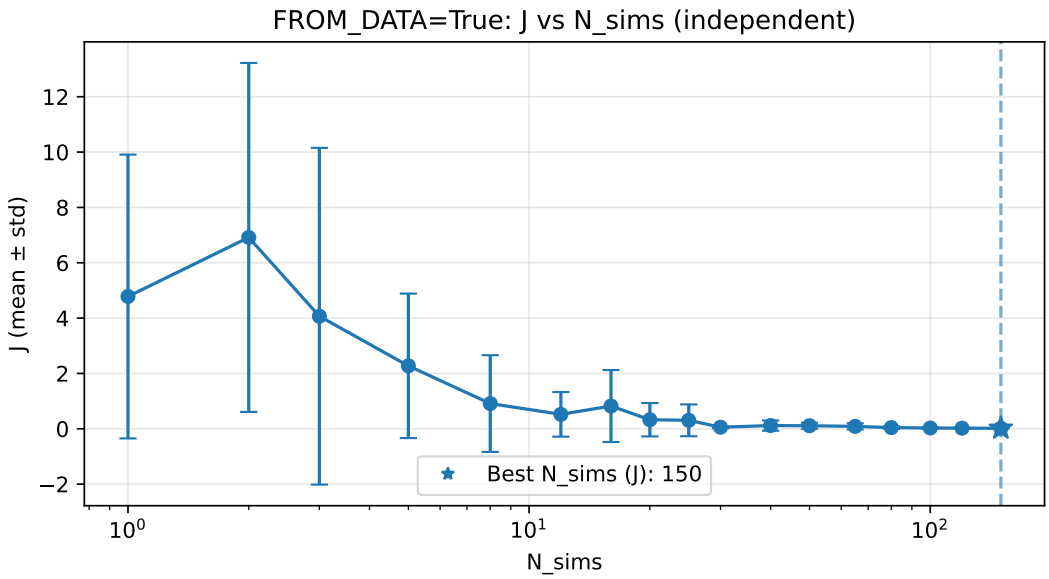


Figure A.4: Closed-loop performance cost J (mean ± std) versus N_{sims} for `FROM_DATA=True` (independent disturbance model). The dashed vertical line and star marker indicate the automatically selected best N_{sims} based on J.

Figures A.4–A.6 summarize these results. The plots show, on a logarithmic horizontal scale, the empirical mean ± one standard deviation for J, `obj`, and $\rho(A_{\text{cl}})$ as N_{sims} increases, together with a vertical dashed line and a star marker highlighting the automatically selected "best" value of N_{sims} for each metric. The J and `obj` curves display a clear variance reduction and gradual decrease as N_{sims} grows, indicating that using more trajectories in the medoid selection improves the robustness of the closed-loop design and makes the cost less sensitive to the particular realization. In contrast, the spectral radius $\rho(A_{\text{cl}})$ stays very close to but below 1 across all values of N_{sims}, with a fairly flat trend: stability is preserved even for small N_{sims}, and increasing the number of trajectories mainly reduces variability rather than shifting the mean far from 1.

Matrix-level analysis across $\mathbf{N_{sims}}$. While `NsimSweep_FROM_DATA` looks at closed-loop scalars, the class `NsimsMatricesAnalyzer` investigates how the identified state-space matrices themselves depend on N_{sims}. For each value in a user-specified list of N_{sims}, the analyzer:

1. builds N_{sims} open-loop datasets, applies the same medoid selection as above and extracts a single representative trajectory;
2. runs the identification pipeline `api.estm_mats` on this trajectory to estimate the state, disturbance, input and performance channels, returning matrices
$$A,\ B_u,\ B_w,\ C_y,\ D_{yw},\ C_z,\ D_{zu},\ D_{zw};$$
3. stacks the resulting estimates over all N_{sims} into arrays of shape $(\#N_{\text{sims}}, \cdot, \cdot)$ and caches them;
4. compares every estimated matrix to the "true" plant matrices
$$A_{\text{true}},\ B_{u,\text{true}},\ B_{w,\text{true}},\ C_{y,\text{true}},\ D_{yw,\text{true}},\ C_{z,\text{true}},\ D_{zu,\text{true}},\ D_{zw,\text{true}}$$

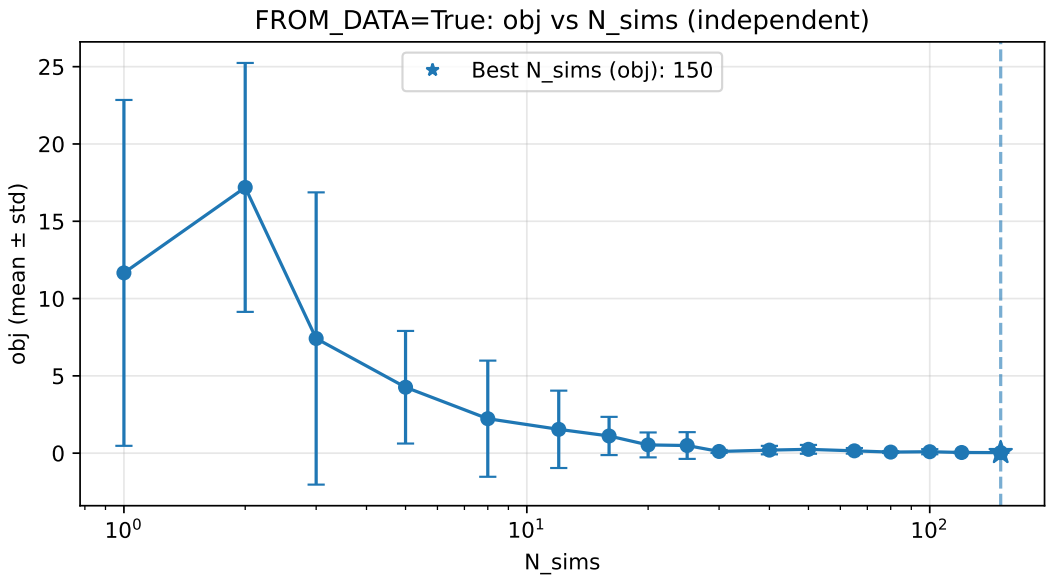


Figure A.5: Optimization objective `obj` (mean ± std) versus N_{sims} under the same setup as in Figure A.4. As more trajectories are used in the medoid selection, the variability of the optimized objective diminishes and the mean value decreases.

obtained from a known model (`FROM_DATA=False`).

On top of the raw storage, the analyzer produces several diagnostic plots. First, for each estimated A it computes the eigenvalues and the spectral radius $\rho(A)$, then plots:

- an overlay of all eigenvalues of A for every N_{sims}, together with the unit circle;
- the spectral radius $\rho(A)$ as a function of N_{sims}.

Second, it computes the step response of the discrete-time system

$$x_{k+1} = A\,x_k + B_u\,u_k, \qquad u_k \equiv 1,$$

from $x_0 = 0$ for a long horizon (here $k = 0, \ldots, 999$), and plots the first state component $x_0[k]$ for each N_{sims} both individually and as an overlay. Finally, for all matrices it computes Frobenius norms, relative Frobenius errors with respect to the true matrices, and heatmaps of entrywise differences, as functions of N_{sims}.

Figure A.7 shows the eigenvalues of A for all N_{sims} on the complex plane, with the unit circle as a stability boundary. All eigenvalues remain inside or extremely close to the unit circle, confirming that the identified closed-loop dynamics stay marginally stable across the entire sweep; the overlays for small and large N_{sims} almost coincide, which indicates that the medoid-based identification is robust to the exact number of trajectories used.

The evolution of the spectral radius $\rho(A)$ with N_{sims} is reported in Figure A.8. All radii cluster extremely close to 1, with small fluctuations of order 10^{-3} across the whole range of N_{sims}. This matches the behaviour seen in Figure A.6: the identified dynamics are consistently near-marginal, and increasing N_{sims} mainly improves the smoothness and reliability of the estimates rather than shifting the spectrum dramatically.

Finally, Figure A.9 shows the step responses of the first state component $x_0[k]$ for all identified models, superimposed in a single panel. All responses are monotone and slowly

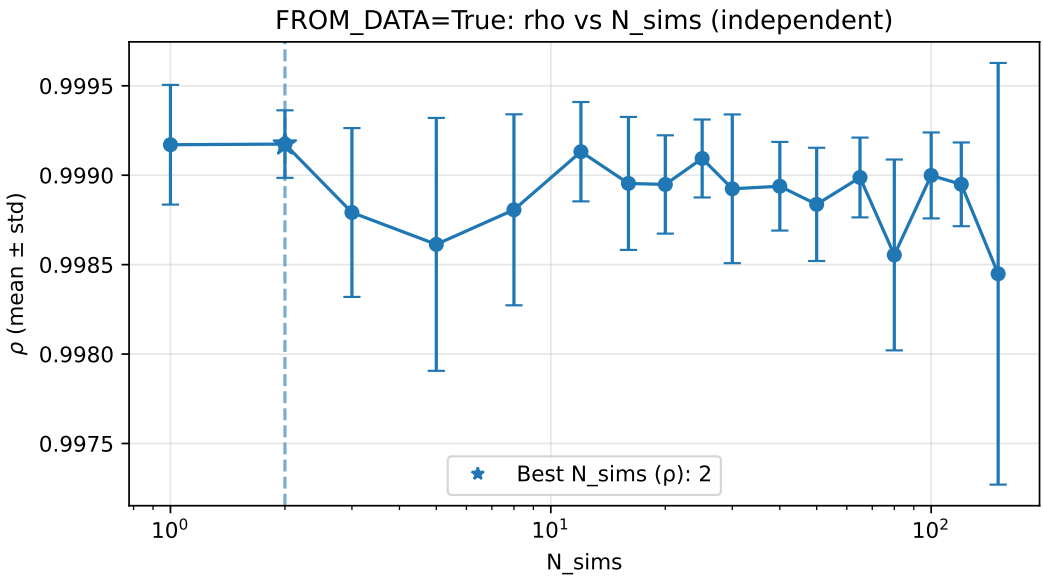


Figure A.6: Spectral radius $\rho(A_{\mathrm{cl}})$ (mean ± std) versus N_{sims}. All designs remain close to the stability boundary $\rho = 1$, but the spread of ρ decreases as N_{sims} increases. The highlighted value corresponds to the best stable N_{sims} according to the selection rule.

decaying, as expected from $\rho(A) \lesssim 1$, but the slope and long-horizon behaviour vary slightly with N_{sims}. The trajectories corresponding to $N_{\mathrm{sims}} = 1$ and $N_{\mathrm{sims}} = 150$ are again highlighted: the high-data regime produces a trajectory that is more representative of the bulk of the overlays, confirming that the medoid computed from many realizations yields matrices whose macroscopic dynamics are more "central" with respect to the ensemble.

Taken together, the scalar performance sweep and the matrix-level analysis show that the medoid selection strategy produces a family of identified models that are spectrally and dynamically consistent across N_{sims}, and that increasing the number of trajectories mostly reduces variance and makes the selected medoid more representative, rather than fundamentally changing the underlying dynamics.

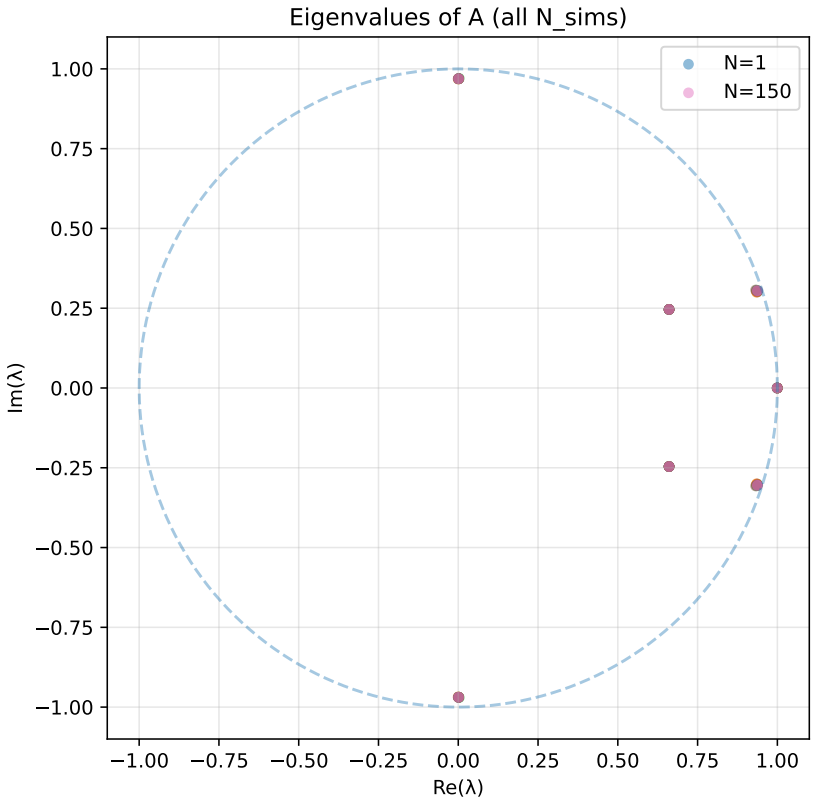


Figure A.7: Eigenvalues of the identified matrix A for all values of N_{sims} (overlay), with the unit circle shown as a dashed line. The points corresponding to $N_{\text{sims}} = 1$ and $N_{\text{sims}} = 150$ are highlighted in the legend.

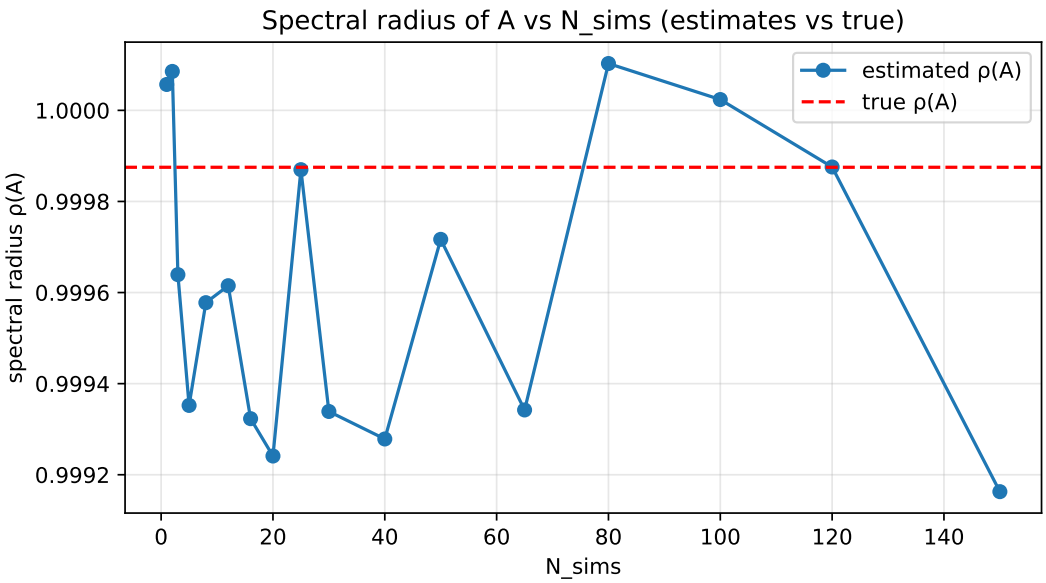


Figure A.8: Spectral radius $\rho(A)$ of the identified state matrix versus N_{sims}, obtained from the matrix-level analyzer. The trend is almost flat and remains very close to 1 for all N_{sims}.

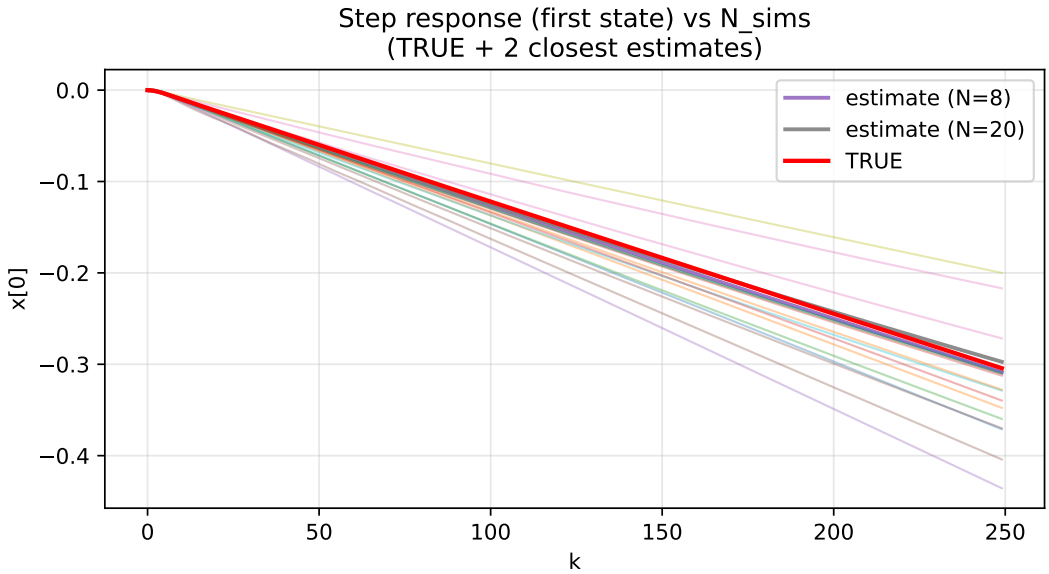


Figure A.9: Step response of the first state component $x_0[k]$ for all identified models, using a unit step input and horizon $k = 0, \ldots, 250$. Curves for $N_{\text{sims}} = 1$ and $N_{\text{sims}} = 150$ are highlighted in the legend.

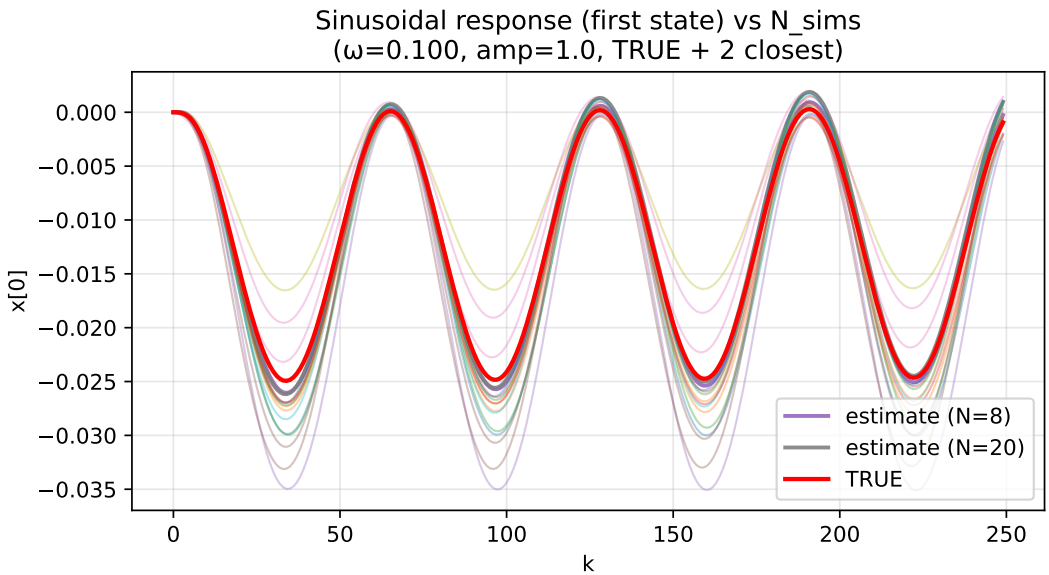


Figure A.10: Sin response of the first state component $x_0[k]$ for all identified models, using a unit step input and horizon $k = 0, \ldots, 250$. Curves for $N_{\text{sims}} = 1$ and $N_{\text{sims}} = 150$ are highlighted in the legend.

Appendix B

Demonstrations

B.1 Closed-loop lifting and decision variables

Recall the following:

$$P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \qquad \begin{aligned} \mathbf{A} &= \begin{bmatrix} \hat{A}Y + \hat{B}_u M & \hat{A} + \hat{B}_u N \hat{C}_y \\ K & X\hat{A} + L\hat{C}_y \end{bmatrix}, \\ \mathbf{B} &= \begin{bmatrix} \hat{B}_w + \hat{B}_u N \hat{D}_{yw} \\ X\hat{B}_w + L\hat{D}_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \hat{C}_z Y + \hat{D}_{zu} M & \hat{C}_z + \hat{D}_{zu} N \hat{C}_y \end{bmatrix}, \\ \mathbf{D} &= \hat{D}_{zw} + \hat{D}_{zu} N \hat{D}_{yw}. \end{aligned}$$

The decision set is $X = X^\top,\ Y = Y^\top,\ K,\ L,\ M,\ N,\ Q,\ \lambda \geq 0$ and the Wasserstein-DRO objective is

$$\min \quad \operatorname{tr}(Q\,\Sigma_{\text{nom}}) + \lambda\gamma^2.$$

Setup. Start from the correlated-noise kernel

$$\Xi(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) := \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I & -Q - \lambda I & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0, \tag{$\star$}$$

with decision variables $P \succ 0$, $Q \succeq 0$, $\lambda \geq 0$, and where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are the usual lifted blocks built from the controller variables (X, Y, K, L, M, N) and the plant maps.

1. **Isolate the Lyapunov generator.** Focus on the $(1,4)$–$(4,1)$ corner that produces the Lyapunov term:

$$M(\mathbf{A}_{\text{true}}, P) := \begin{bmatrix} -P & \mathbf{A}_{\text{true}}^\top \\ \mathbf{A}_{\text{true}} & -P \end{bmatrix} \prec 0.$$

Starting from the Schur complement of the $(4,4)$ block $-P$ we obtain

$$-P - \mathbf{A}_{\text{true}}^\top (-P)^{-1} \mathbf{A}_{\text{true}} = -P + \mathbf{A}_{\text{true}}^\top P^{-1} \mathbf{A}_{\text{true}} \prec 0. \tag{1}$$

This form expresses discrete-time stability in the inverse-P coordinates.

2. **Inject identification error.** Let $A = \hat{A} + \Delta_A$, $B_u = \hat{B}_u + \Delta_B$, with data-driven bounds $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$. In the lifted block, $\mathbf{A}$ depends *affinely* on (A, B_u), so replacing by $\hat{A}, \hat{B}_u$ plus errors gives

$$\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E_A + E_B,$$

where E_A collects the contributions linear in Δ_A, and E_B those linear in Δ_B. In particular:

$$E_A = \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix} \qquad E_B = \begin{bmatrix} \Delta_B M & \Delta_B N C_y \\ 0 & 0 \end{bmatrix} \tag{B.1}$$

Expanding (L) with $\mathbf{A}_{\text{true}}$, with $E = E_A + E_B$, gives

$$\mathbf{A}^\top P^{-1}\mathbf{A} - P \;+\; \underbrace{\mathbf{A}^\top P^{-1}E + E^\top P^{-1}\mathbf{A}}_{\text{mixed terms}} \;+\; \underbrace{E^\top P^{-1}E}_{\text{2nd order}} \prec 0,$$

where $\mathbf{A} := \mathbf{A}(\hat{A}, \hat{B}_u)$ is the nominal lifted block.

3. **Bound on the perturbation block E.** We start from the base bound

$$\beta = RD^\dagger \in \mathbb{R}^{(n_x+n_u)\times(n_x+n_u)},$$

and split it between the state and input parts as

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x+n_u}}, \qquad \beta_B = \beta\sqrt{\frac{n_u}{n_x+n_u}}.$$

The corresponding spectral–norm bounds are

$$\kappa_A := \|\beta_A\|_2, \qquad \kappa_B := \|\beta_B\|_2,$$

and we assume

$$\|\Delta_A\|_2 \le \kappa_A, \qquad \|\Delta_B\|_2 \le \kappa_B.$$

The lifted perturbation block is

$$E \;=\; \begin{bmatrix} \Delta_A Y + \Delta_B M & \Delta_A + \Delta_B N C_y \\ 0 & X\Delta_A \end{bmatrix},$$

where the design matrices satisfy the norm bounds

$$\|Y\|_2 \le t_Y, \qquad \|M\|_2 \le t_M, \qquad \|N\|_2 \le t_N, \qquad \|X\|_2 \le t_X,$$

and C_y is constant with $c_y := \|C_y\|_2$.

Using submultiplicativity and the triangle inequality, we obtain

$$\|\Delta_A Y + \Delta_B M\|_2 \le \|\Delta_A\|_2\|Y\|_2 + \|\Delta_B\|_2\|M\|_2 \le \kappa_A t_Y + \kappa_B t_M =: e_{11},$$

$$\|\Delta_A + \Delta_B N C_y\|_2 \le \|\Delta_A\|_2 + \|\Delta_B\|_2\|N\|_2\|C_y\|_2 \le \kappa_A + \kappa_B t_N c_y =: e_{12},$$

$$\|X\Delta_A\|_2 \le \|X\|_2\|\Delta_A\|_2 \le t_X\kappa_A =: e_{22}.$$

A conservative global bound on E follows from

$$\|E\|_2^2 \;\le\; \|E\|_F^2 \;\le\; e_{11}^2 + e_{12}^2 + e_{22}^2 =: \gamma_E,$$

which implies

$$E^\top E \;\preceq\; \beta_E \quad \text{with} \quad \beta_E := \gamma_E I = \big(e_{11}^2 + e_{12}^2 + e_{22}^2\big) I.$$

This constant matrix β_E is then used as an upper bound for the perturbation energy in the subsequent LMIs.

4. Robustification of the Lyapunov inequality under model error.

Bounding the inverse metric. Introduce $\sigma_p > 0$ such that

$$P^{-1} \prec \sigma_p I. \tag{3.2}$$

This is equivalent to

$$\begin{bmatrix} \sigma_p I & I \\ I & P \end{bmatrix} \succ 0. \tag{3.3}$$

Thus

$$E^\top P^{-1} E \preceq \sigma_p E^\top E, \qquad E^\top E \preceq B_E \quad \Rightarrow \quad E^\top P^{-1} E \preceq \sigma_p B_E. \tag{3.5}$$

Bounding the mixed term. For any $S \succ 0$,

$$E^\top P^{-1}\mathbf{A} + \mathbf{A}^\top P^{-1} E = 2\langle E,\; P^{-1}\mathbf{A}\rangle \;\le\; E^\top S E + \mathbf{A}^\top P^{-1} S^{-1} P^{-1}\mathbf{A}. \tag{3.6}$$

Introduce $\sigma_s > 0$ with

$$S \prec \sigma_s I \quad \Longleftrightarrow \quad \sigma_s I - S \succ 0. \tag{3.7}$$

Then

$$E^\top S E \preceq \sigma_s B_E. \tag{3.8}$$

Combine all bounds. Using (3.5), (3.6), (3.8):

$$-P + \mathbf{A}^\top\big(P^{-1} - P^{-1} S P^{-1}\big)\mathbf{A} + \sigma_p B_E + \sigma_s B_E \prec 0. \tag{3.9}$$

Define

$$\bar{P} := P - \sigma_e B_E, \quad \sigma_e = \sigma_s + \sigma_p, \tag{3.10}$$

so $P_s \succ 0$. Its inverse satisfies

$$P_s^{-1} = P^{-1} + P^{-1} S P^{-1} \succ 0 \tag{3.11}$$

Therefore

$$-\bar{P} + \mathbf{A}^\top P_s^{-1}\mathbf{A} \prec 0 \quad \Longleftrightarrow \quad \begin{bmatrix} -\bar{P} & \mathbf{A}^\top \\ \mathbf{A} & -P_s \end{bmatrix} \prec 0. \tag{3.12}$$

To bound P_s^{-1} we do the following.

Auxiliary lifting. Define

$$G = \sigma_p I = G^\top \succ 0, \qquad G \succ P^{-1}, \tag{3.13}$$

and introduce slack $H \succ 0$ via

$$\begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0 \quad \Longleftrightarrow \quad H - GS^{-1}G \succ 0. \tag{3.15}$$

Final condition. The robust requirement becomes

$$G + H - P_s^{-1} \succ 0, \tag{3.16}$$

equivalently,

$$\begin{bmatrix} G - H & I \\ I & P_s \end{bmatrix} \succ 0. \tag{3.17}$$

And thus, $P_s^{-1} = P^{-1} + P^{-1}S^{-1}P^{-1} \prec G + GS^{-1}G \prec G + H$.

5. **Regularization term (convex surrogate).**

To control the operator norms of the learned matrices K, L, M, N, introduce auxiliary scalar variables $t_K, t_L, t_M, t_N \geq 0$ and impose the epigraph LMIs

$$\begin{bmatrix} t_K I_{n_x} & K \\ K^\top & t_K I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_L I_{n_x} & L \\ L^\top & t_L I_{n_y} \end{bmatrix} \succeq 0, \tag{B.2}$$

$$\begin{bmatrix} t_M I_{n_u} & M \\ M^\top & t_M I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_X I_{n_x} & X \\ X^\top & t_X I_{n_x} \end{bmatrix} \succeq 0, \tag{B.3}$$

$$\begin{bmatrix} t_Y I_{n_x} & Y \\ Y^\top & t_Y I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_N I_{n_u} & N \\ N^\top & t_N I_{n_y} \end{bmatrix} \succeq 0. \tag{B.4}$$

These LMIs ensure $\|K\|_2 \leq t_K$, $\|L\|_2 \leq t_L$, $\|X\|_2 \leq t_X$, $\|Y\|_2 \leq t_y$, $\|M\|_2 \leq t_M$, and $\|N\|_2 \leq t_N$.

A convex spectral-norm regularization term is then added to the objective:

$$\mathcal{R}_{\text{spec}} = \rho\,(t_K + t_L + t_M + t_N + t_X + t_Y),$$

where the weights $\rho > 0$ tune the relative penalization of each block.

Optionally, a small numerical offset $\varepsilon > 0$ can be included to improve conditioning, yielding

$$\mathcal{R}_{\text{spec},\varepsilon} = \rho\,[(t_K + \varepsilon) + (t_L + \varepsilon) + (t_M + \varepsilon) + (t_M + \varepsilon) + (t_X + \varepsilon) + (t_Y + \varepsilon)].$$

Together with the previous convex surrogate term $\mathcal{R}_\varepsilon$, these define the complete regularized optimization problem, with $\mathcal{R} = \mathcal{R}_\varepsilon + \mathcal{R}_{\text{spec},\varepsilon}$:

$$\min_{\text{decision vars}} \; J_{\text{main}} + \mathcal{R} \quad \text{s.t.} \quad \text{LMI constraints above.}$$

B.2 Post-optimization recovery of Δ_A and Δ_B

Inputs. After solving the program, suppose we have optimal quantities

$$P^\star \succ 0, \quad X^\star = X^{\star\top}, \quad Y^\star = Y^{\star\top}, \quad M^\star,\ N^\star,\ s_A^\star > 0,\ s_B^\star > 0,\ \tau_A^\star,\ \tau_B^\star,$$

together with nominal identified maps $\hat{A}$, $\hat{B}_u$, $\hat{C}_y$ and data-driven radii β_A, β_B that induce the lifted bounds

$$\|E_A\|_F \le \beta_A := \sqrt{1 + \|X^\star\|_2^2 + \|Y^\star\|_2^2}\,\beta_A, \qquad \|E_B\|_F \le \beta_B := \sqrt{\|M^\star\|_2^2 + \|N^\star\|_2^2\|\hat{C}_y\|_2^2}\,\beta_B.$$

Step 1: Adversarial directions for the Young bounds. Let the nominal lifted block (built with $\hat{A}, \hat{B}_u$) be

$$\mathbf{A}^\star := \mathbf{A}(\hat{A}, \hat{B}_u; X^\star, Y^\star, M^\star, N^\star), \qquad Z_{A,i}^\star := P^\star \mathbf{A}^\star, \quad i \in \{A, B\}.$$

Choose worst-case aligned perturbations that saturate the Young relaxation:

$$E_{A,i}^\star = \begin{cases} \beta_{A,i} \dfrac{Z_{A,i}^\star}{\|Z_{A,i}^\star\|_F}, & Z_{A,i}^\star \neq 0, \\ 0, & Z_{A,i}^\star = 0, \end{cases} \qquad \beta_{A,A} := \beta_A, \quad \beta_{A,B} := \beta_B. \tag{B.5}$$

This choice attains equality in Young's inequality when $s_{A,i}^\star = \|Z_{A,i}^\star\|_F / \beta_{A,i}$.

Step 2: Block structure of $E_A^\star$ and $E_B^\star$. Recall the definitions

$$E_A = \begin{bmatrix} \Delta_A Y^\star & \Delta_A \\ 0 & X^\star \Delta_A \end{bmatrix}, \qquad E_B = \begin{bmatrix} \Delta_B M^\star & \Delta_B N^\star \hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

After choosing $E_A^\star, E_B^\star$ via (B.5), extract the nonzero sub-blocks

$$A_1 := E_A^\star(1{:}n,\ 1{:}n), \quad A_2 := E_A^\star(1{:}n,\ n{+}1{:}2n), \quad A_3 := E_A^\star(n{+}1{:}2n,\ n{+}1{:}2n),$$

$$B_1 := E_B^\star(1{:}n,\ 1{:}m), \qquad B_2 := E_B^\star(1{:}n,\ m{+}1{:}m{+}p),$$

Step 3: Recovery of Δ_A (Sylvester equation). Solve the least-squares normal equation

$$\boxed{(X^{\star\top} X^\star)\,\Delta_A \;+\; \Delta_A\,(Y^\star Y^{\star\top} + I) \;=\; A_2 + A_1 Y^{\star\top} + X^{\star\top} A_3\ ,} \tag{B.6}$$

which is a Sylvester equation in Δ_A. Under mild conditions on the spectra of $X^{\star\top} X^\star$ and $Y^\star Y^{\star\top} + I$, (B.6) has a unique solution.

Special cases. If $X^\star = 0$, then $\Delta_A(Y^\star Y^{\star\top} + I) = A_2 + A_1 Y^{\star\top} \Rightarrow \Delta_A = (A_2 + A_1 Y^{\star\top})(Y^\star Y^{\star\top} + I)^{-1}$. If $Y^\star = 0$, then $(X^{\star\top} X^\star + I)\Delta_A = A_2 + X^{\star\top} A_3 \Rightarrow \Delta_A = (X^{\star\top} X^\star + I)^{-1}(A_2 + X^{\star\top} A_3)$.

Step 4: Recovery of Δ_B (right-side Gram solve). Solve

$$\boxed{\Delta_B\, G_B \;=\; B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top}, \qquad G_B := M^\star M^{\star\top} + N^\star \hat{C}_y \hat{C}_y^\top N^{\star\top}\ ,} \tag{B.7}$$

so that, if G_B is invertible, $\Delta_B = (B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top})\, G_B^{-1}$. If G_B is singular or ill-conditioned, use the Moore–Penrose pseudoinverse G_B^+ or a Tikhonov-regularized inverse $(G_B + \eta I)^{-1}$ with small $\eta > 0$.

Step 5: Augmented nominal maps. Finally set

$$A_{\text{aug}} = \hat{A} + \Delta_A, \qquad B_{u,\text{aug}} = \hat{B}_u + \Delta_B,$$

and, if desired, verify feasibility with the identification radii: $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$ and with the lifted bounds $\|E_A(\Delta_A)\|_F \le \beta_A$, $\|E_B(\Delta_B)\|_F \le \beta_B$.

Remarks. (i) The recovered (Δ_A, Δ_B) are *representative* worst-case directions induced by the Young relaxation, not the unique identification errors. (ii) Numerically, (B.6) is solved with a Sylvester solver; (B.7) is a right-multiplication by a Gram factor and should use a robust linear solver or SVD-based pseudoinverse. (iii) If a single uncertainty block on $[A\ B_u]$ was used in synthesis, replace β_A, β_B by a common β and set $E_x^\star = \beta\, Z_x^\star / \|Z_x^\star\|_F$ with $Z_x^\star = P^\star[\hat{A}\ \hat{B}_u]$, then project $E_x^\star$ onto the E_A, E_B structures above before solving (B.6)–(B.7).

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